

SAULT STE MARIE, ON, Canada

WMO: 712600

Lat: 46.4840N

Lon: 84.4992W

Elev: 192

StdP: 99.04

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.2	-22.0	-29.9	0.2	-24.8	-26.4	0.3	-21.4	12.9	-4.8	11.1	-6.3	1.5	80	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	28.5	21.1	26.8	20.0	25.2	19.1	22.3	26.6	21.2	25.2	20.2	23.9	3.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	15.9	24.8	19.8	14.9	23.4	18.7	13.9	22.1	66.5	26.7	62.3	25.1	58.8	23.9	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.9	7.9	-30.0	31.1	4.0	1.4	-32.9	32.1	-35.3	32.9	-37.5	33.7	-40.5	34.8		
(5)			-30.2	24.3	3.9	1.1	-33.0	25.1	-35.4	25.8	-37.6	26.4	-40.4	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-9.5	-8.7	-4.0	3.2	10.2	15.4	18.2	17.9	14.4	7.7	1.4	-4.8	(6)
(7)		DBStd	10.99	6.83	6.39	6.37	4.51	4.33	3.76	3.15	2.93	4.11	4.34	4.81	5.66	(7)
(8)		HDD10.0	2653	603	524	436	208	52	3	0	0	8	99	260	459	(8)
(9)		HDD18.3	4896	862	757	693	454	256	102	42	44	130	329	509	717	(9)
(10)		CDD10.0	896	0	0	1	4	56	166	254	244	140	29	1	0	(10)
(11)		CDD18.3	98	0	0	0	0	3	15	38	30	12	1	0	0	(11)
(12)		CDH23.3	816	0	0	1	1	41	167	328	204	72	2	0	0	(12)
(13)		CDH26.7	153	0	0	0	0	7	33	71	32	11	0	0	0	(13)
(14)	Wind	WSAvg	3.4	3.5	3.6	3.6	3.8	3.6	3.1	2.9	2.8	2.9	3.6	3.9	3.8	(14)
(15)	Precipitation	PrecAvg	932	78	48	61	69	70	81	74	86	100	87	93	84	(15)
(16)		PrecMax	1228	128	140	136	120	186	230	165	174	201	158	175	161	(16)
(17)		PrecMin	632	23	6	12	30	15	10	8	24	33	8	30	11	(17)
(18)		PrecStd	130	25	27	30	26	35	41	35	32	42	34	34	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	5.8	13.9	20.3	27.5	29.8	30.8	29.5	28.3	22.8	15.1	7.6	(19)
(20)			MCWB	4.0	3.9	9.4	12.4	19.3	21.9	22.3	22.0	21.4	17.8	11.7	6.5	(20)
(21)		2%	DB	2.7	3.2	9.2	16.2	23.8	27.2	28.6	27.4	25.3	19.3	11.5	5.2	(21)
(22)			MCWB	1.8	1.7	5.6	9.4	15.9	19.9	21.2	20.8	19.5	15.6	9.3	4.1	(22)
(23)		5%	DB	1.4	1.5	6.2	13.1	21.0	25.3	26.8	25.7	23.2	16.6	9.5	3.7	(23)
(24)			MCWB	0.5	0.4	3.4	7.4	14.4	18.4	20.0	19.8	18.5	13.9	7.9	2.8	(24)
(25)		10%	DB	0.1	0.2	3.9	10.3	18.1	23.0	25.0	23.9	21.2	14.4	7.8	2.4	(25)
(26)			MCWB	-0.7	-0.8	1.9	6.1	12.5	17.1	19.2	18.7	17.4	12.2	6.4	1.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	4.3	10.3	13.5	20.0	23.2	24.0	23.2	22.1	18.7	12.4	6.9	(27)
(28)			MCDWB	4.7	5.3	12.4	18.7	26.3	28.7	28.8	27.6	26.4	20.7	14.4	7.3	(28)
(29)		2%	WB	2.0	1.8	6.4	10.3	17.6	21.1	22.4	21.9	20.6	16.2	9.8	4.3	(29)
(30)			MCDWB	2.6	2.8	8.7	14.3	22.3	26.0	26.9	25.6	23.9	18.7	11.1	4.9	(30)
(31)		5%	WB	0.7	0.6	3.8	8.1	15.2	19.5	21.1	20.8	19.4	14.2	8.0	2.9	(31)
(32)			MCDWB	1.3	1.5	5.6	12.4	19.1	23.8	25.4	24.2	22.2	16.2	9.1	3.6	(32)
(33)		10%	WB	-0.7	-0.8	2.0	6.3	13.3	18.0	20.0	19.8	18.1	12.5	6.5	1.6	(33)
(34)			MCDWB	0.1	0.1	3.8	9.7	17.3	21.6	23.8	22.9	20.7	14.3	7.8	2.3	(34)
(35)	Mean Daily Temperature Range	MDBR	8.5	9.2	9.8	9.6	11.6	11.8	11.3	10.6	10.1	8.0	6.4	6.8	(35)	
(36)		5% DB	MCDBR	6.5	8.0	10.8	15.5	16.5	15.5	14.6	13.2	12.4	11.2	8.7	6.8	(36)
(37)			MCWBR	6.2	7.1	7.7	9.7	9.7	8.9	8.0	7.4	7.3	7.8	7.0	6.3	(37)
(38)			5% WB	MCDBR	6.3	7.8	9.6	13.6	14.2	13.1	12.2	10.7	10.9	9.9	7.7	6.7
(39)		MCWBR		6.1	7.2	7.2	9.3	9.2	8.2	7.3	6.5	7.1	7.3	6.7	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.280	0.290	0.319	0.356	0.370	0.388	0.389	0.379	0.366	0.331	0.302	0.286	(40)	
(41)		taud	2.339	2.297	2.330	2.330	2.374	2.345	2.363	2.392	2.424	2.504	2.508	2.373	(41)	
(42)		Ebn at Noon	817	883	907	900	895	877	871	867	847	831	794	765	(42)	
(43)		Edh at Noon	74	94	107	118	117	121	118	110	98	78	64	65	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.07	1.99	3.25	4.45	5.56	5.99	6.03	5.10	3.80	2.21	1.15	0.83	(44)	
(45)		RadStd	0.11	0.21	0.37	0.51	0.45	0.40	0.34	0.34	0.28	0.25	0.15	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page